CPE 314 — Software Design

University of Cebu

Computer Engineering Department

Pet Shelter & Adoption Tracker

Project Design Document

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Revision History

Update this table each week as your team adds or revises sections of this document.

Version	Date	Author	Changes
0.1	Oct 21, 2026	Group 3	Initial draft — Week 8 sections
0.2			Week 9 sections added
0.3			Week 10 sections added
1.0			Final compilation and polish — Week 11

Formatting Guidelines

Follow these formatting rules consistently throughout the document so the final compiled version reads as one polished deliverable, not four stitched-together drafts.

Element	Rule
Body font	Calibri, 11pt (the default font of this template — do not mix fonts)
Headings	Use Word's built-in Heading 1/2/3 styles (already applied to every section title below) — this is required for the Table of Contents field to work
Margins	1 inch on all sides
Line spacing	1.15–1.5 line spacing for body paragraphs; single-spacing inside tables
Table captions	Label every table "Table N: short description" directly ABOVE the table, numbered sequentially from Table 1 through the end of the document
Figure/chart captions	Label every chart/screenshot "Figure N: short description" directly BELOW the figure, numbered sequentially from Figure 1
Code snippets	Use a monospace font (Consolas or Courier New), 9–10pt, and keep each snippet under ~20 lines — link to the full file in your repository instead of pasting it all
Page numbers	Already automatic — every page footer includes a live page number field
Alignment	Body text left-aligned/justified; headings left-aligned; do not center body paragraphs

1. Introduction & Project Overview

The Pet Shelter & Adoption Tracker (Option P) is a Java desktop application designed to help animal shelter staff manage intake, waitlists, and adoptions in one offline-first tool. Many small shelters rely on scattered spreadsheets and paper records, making it difficult to track animal stays and adoption demand. Our application centralizes these operations: it models species with distinct care schedules, manages adoption waitlists using queues, enables status undo operations via a stack, and supports bulk historical CSV log imports. Integrated pet APIs (Dog CEO API and TheCatAPI) auto-fill sample images and breed details, while built-in analytics render adoption trends and stay durations as visual bar charts to assist staff in operational planning.

2. Preliminary ER Sketch & Team Roles

Insert the preliminary entity-relationship sketch from your Week 8 proposal — entities, key attributes, and at least one relationship — and list every team member's name and primary role. This is a rough draft; it will be superseded by the finalized ERD in Section 4.

Table 1: Preliminary entities and key attributes

Entity	Key Attributes
Classification	id (PK), name, parent_id (FK → Classification.id)
Animal	id (PK), name, species, breed, age_months, intake_date, status, image_url classification_id (FK → Classification.id)
Adopter	id (PK), name, email (unique), phone
AdoptionRecord	id (PK), animal_id (FK → Animal.id), adopter_id (FK → Adopter.id), adoption_date, fee_paid
WaitlistEntry	id (PK), animal_id (FK → Animal.id), adopter_id (FK → Adopter.id), requested_at, position

Table 2: Preliminary relationships and cardinality

Parent Entity	Relationship	Child Entity	Foreign Key	Description
Classification	1:M	Classification	parent_id (nullable)	Self-referencing taxonomy tree of any depth, e.g. Dog > Sporting Group > Retriever > Golden Retriever
Classification	1:M	Animal	classification_id	Each animal belongs to one classification
Animal	1:M	AdoptionRecord	animal_id	An animal can have adoption records
Adopter	1:M	AdoptionRecord	adopter_id	An adopter can adopt multiple animals
Animal	1:M	WaitlistEntry	animal_id	An animal can have many adopters waiting
Adopter	1:M	WaitlistEntry	adopter_id	An adopter can wait on multiple animals

Table 3: Team members and primary roles	
Member	**Primary Responsibility (this milestone)**
Sofia Yvonne Enovero	OOD model + class hierarchy, Queue/Stack, recursive tree search
Shann Michael Agang	Swing GUI + event handlers
Cleint Lausa	SQLite / JDBC data layer & DAOs
Kyle Cheznie Taghoy	REST API client + JSON parsing + CSV parsing/validation
Aeisha Mae Elizar	Concurrency (import/API workers) + analytics & chart

Note: Each person writes tests for their own module.

3. System Architecture & Object-Oriented Design

Present your class hierarchy (a diagram or a clear textual description) showing the abstract class/interface and its concrete subclasses, and justify how each class follows the Single Responsibility Principle. Explain the design rationale in a short paragraph.

4. Data Model & Database Design

Include the finalized ER diagram (3+ entities, at least one relationship with correct cardinality) and the actual SQLite schema (CREATE TABLE statements or an equivalent schema table). This section supersedes the preliminary sketch from Section 2.

5. API Integration Design

Name the public API your app consumes, describe the endpoint(s) called, the shape of the JSON response, and how that data is parsed and used meaningfully in the app.

6. Concurrency Design

Identify the long-running operation(s) that run on a background thread, explain the threading approach used, and describe how the UI stays responsive during that operation.

7. Data Mining & Visualization Design

Describe the aggregation or frequency-analysis step performed on your app's data and the chart(s) used to visualize the result, including why that chart type was chosen.

8. Testing Summary

Summarize your JUnit test suite — how many tests exist, what they cover, and at least one meaningful edge case per tested class. A short table or list of key test cases is helpful here.

9. Individual Contributions

List what each team member specifically built or owned, referencing their git commit history, so the instructor can cross-check contribution during the Individual Q&A.

Appendix

(Add any additional references, links to full API documentation, licensing notes, or supplementary material here.)